

Exploring Public Converse on AI via Reddit: A Topic Modeling and Sentiment Analysis Approach

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Abstract: The 21st Century is a compressed century, where Artificial Intelligence (AI) plays a crucial role in the new phase of digital transformation, disrupting various aspects of society, including art, education, and healthcare. Throughout this process, the public has responded with multifarious perspectives. Understanding these viewpoints can help foster responsible AI development by analyzing both its benefits and challenges. While many current studies focus solely on products from OpenAI, such as ChatGPT, this study takes a more comprehensive approach. Data were collected via Application Programming Interface (API) from AI-related subreddits on Reddit, one of the largest social platforms in the United States. This dataset was used to train three topic modeling algorithms: Latent Dirichlet Allocation (LDA), Non-negative Matrix Factorization (NMF), and Bertopic. After comparing these models based on various metrics, topic representation was fine-tuned using a semi-AI approach. Subsequently, a high-dimensional analysis was conducted through techniques such as sankey diagrams, dynamic topic modeling, and sentiment analysis. The results reveal online concerns regarding AI and this study further analyzes users' behaviors and discusses broader implications.

Key words: artificial intelligence; natural language processing; topic modeling; sentiment analysis; Bertopic; Reddit

I Introduction

Artificial Intelligence (AI) has a significant impact on the entire human society. One of the most well-known AI techniques is Generative AI, which disrupts the traditional framework of education, art, and any other fields that require creation. Other examples would be the Artificial Intelligence Internet of Things, where the participation of AI makes edging devices

smarter than ever, such as smart homes, smart healthcare, and intelligent industries^[1]. At the same time, the research regarding AI is also increasing in different areas, such as expert systems, machine learning, and robotics^[2]. Meanwhile, the fast development of AI also leads to ethical, security, and safety concerns. Although AI can play a critical role in detecting crimes by integrating various techniques^[3], AI such as Generative AI can also be used as a weapon to generate malicious content via jailbreaking, reverse psychology, or other techniques^[4]. Additionally, deep learning techniques, which are closely related to AI, can create potential problems such as discrimination and bias, leading to negative social impact^[5].

Moreover, the workforce market is also impacted by AI, where using AI to unleash productivity is becoming more common. As Ref. [6] revealed, AI can effectively improve low-skill workers' productivity. Reference [7]

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inferred that workers who are not innovative would face the risk of job displacement. Furthermore, fear of Generative AI towards safety is growing. Indeed, this concern is not unwarranted. Reference [8] proposed that deepfake content facilitated by AI diminishes public morale, and this phenomenon can be aggravated if the audience has low cognitive abilities. As a result, although AI shows its effectiveness and power in handling numerous tasks, it is not without its drawbacks. In other words, AI has to be human-centered and must avoid and ameliorate anything that is against well-being. Therefore, while the development and publication of new AI techniques receive tremendous attention, public perspective towards AI is also crucial.

Topic modeling is a widely used technique to extract themes from a collection of documents, facilitating the process of understanding a target group's viewpoints^[9]. Social media plays a critical role as a data source in topic modeling. For instance, social media can provide valuable insight of consumers for markets to improve their strategy^[10]. In the other example, researchers can incorporate sentiment analysis and topic modeling on social media data to predict stock price efficiently^[11]. In this study, Reddit, as one of the biggest social platforms in the United States and also a valuable source for topic modeling studies, was selected as the data source. Benefiting from Reddit's large user base, which is over 500 million Reddit accounts, various research has been conducted on it, exploring healthcare, education, AI, and other aspects of society. While some investigations were undertaken regarding AI, they mainly focused on OpenAI's products, such as ChatGPT and Sora. This study aims to fill this research gap by collecting a more diverse AI-related textual dataset on Reddit and providing a more comprehensive and in-depth analysis. More specifically, this study will make the following contributions.

- Extend the scope of AI-related research on Reddit by collecting data from various subreddits that focus on different areas related to AI.
- Produce a more interpretable and insightful result of how the public reacts to different aspects of AI by fine-tuning the topic representation through a semi-AI approach.
- Classified topics into fields, and provide a high dimensional analysis through composition, time, and sentiment.

The rest of the study is organized as follows: Section 2 makes an inventory of recent papers that applied topic modeling on Reddit data; Section 3 is divided into four parts, which are data obtained from Reddit, topic modeling models and optimization techniques used, evaluation metrics used to measure the performance of the models, and topic fine-tuning technique that is employed; Section 4 firstly compares the models, then gives an in-depth analysis of best model's topics from multiple aspects. Finally, in Section 5, a summary and future steps of this study are discussed.

2 Background and Related Work

The existing research on topic modeling on Reddit had explored a variety of perspectives and approaches. This section synthesized the findings from 19 key papers in the field of health, AI, and social phenomenon and social dynamics. The collection process of these papers was through Google Scholar, where keywords such as "topic modeling", "Reddit", "health", and "impact" are used in the search query. With a focus on recent studies, the published date of selected studies ranged from 2020 to 2024. Table 1 shows the selected study in the corresponding category.

Among these three areas, health has been the most discussed. Reference [12] examined the influence of toxic behavior from surrounding people to breast cancer patients, such as the impact from partners and family members. Following the findings in Ref. [12], Ref. [13] explored another aspect of physical health, investigating the public's concern about endometriosis the data from two related subreddits and classified them into different themes using the model LDAMulticore. Reference [13] further analyzed the sentiment of their data through sentiment analysis, emphasizing the increased need for attention and support for women who suffer from endometriosis.

Whlist Ref. [14] also focuses on a physical health disease, cystic fibrosis, it incorporated time-series

Table 1 Summary of related works.

Category	Reference
Health	[12–18]
Health, AI	[19]
AI	[20–22]
Social phenomena, AI	[23, 24]
Social phenomena	[25–30]

analysis and Bertopic to study the trend in the cystic fibrosis community under the COVID-19. Moving the view from physical health to mental health, Ref. [15] focused on how applied behavioral analysis can be an effective treatment for Autism Spectrum Disorder. While also studying Spectrum Disorder, Ref. [16] shifted the view to Alzheimer's disease and related dementia, inquiring into identifying the most prevalent discussion or advice that occurred among the caregivers using Bertopic. In Ref. [17], the authors studied Maternal Health using latent Dirichlet Analysis, providing insights of the health of women during pregnancy, childbirth, and the postnatal period. Reference [18] shed light on weight loss, where the authors performed LDA to extract topics from the post data, which was created by the Reddit users who recorded their weight changes. The authors further applied regression analysis to learn the relationship between topics and the actual weight loss.

As AI becomes more and more prevalent, several studies have been conducted recently around it. ChatGPT is one of the most-used AI tools around the world, and Ref. [20] classified users' interaction with ChatGPT into different themes and applied sentiment analysis and time-series analysis to study the trend of their opinions on ChatGPT. Reference [23] examined ChatGPT's potential in the education area by using LDA and sentiment analysis to analyze users' experience using ChatGPT for educational purposes. In addition, the research in Ref. [24] delves into the impact ChatGPT brings to streaming media. This study debated the pros and cons of ChatGPT in streaming media, and divulged users' perspectives from various dimensions.

Another service from Open AI also gained attention, the research in Ref. [21] analyzed the public perception of Sora, a text-to-video model. The authors utilized LDA to extract topics from subreddit comments, disclosing public perception as well as artists' perception towards Sora, assessing the trend of Sora, and talking about Sora's application in media and entertainment. In addition, Ref. [22] explored the safety concern raised by generated AI due to the increase in cases of deepfakes, uncovering communities' attitudes towards deepfakes and mooting the implication of such phenomenon. Moreover, Ref. [19] contributed to both AI and health, which discovered

the public's sentiment towards ChatGPT, and discussed the potential of ChatGPT in mental health counseling.

Topic modeling can also provide valuable insights into social phenomena and social dynamics. For instance, education issues are frequently discussed. Both Refs. [25, 26] discovered the concerns among educators and learners behind the online lessons under the COVID-19, where LDA is employed on data collected from various subreddits, while Ref. [25] further performed sentiment analysis and time-series analysis to uncover the trend of public emotion and attention toward online classes. Shifting the center from online classroom to family, Ref. [27] investigated potential issues that parents could bring to their children by using LDA to extract themes from two selected subreddit, assisting the development of more effective parenting methods. Except for education, there are several studies on online behaviors.

Reference [28] probed toxic behaviors on Reddit, such as hate speech and cyberbullying, and measured the "health" of communities by examining the relationship between toxicity and users' engagement. Reference [29] deepened the analysis of emotional eating behavior, an eating disorder affected by emotion, by employing LDA to identify highly associated topics to this syndrome. In addition, the technology topic also gained attention, where Ref. [30] inspected public perception of Electronic Vehicles (EV), revealing community-favored EV-related topics and their trends.

3 Methodology

3.1 Data source

Social media fosters a pluralist environment for discussion, facilitating delivery and exchange of ideas across people from diverse social milieu. Reddit, one of the biggest social platforms in the United States with over 500 million accounts, enables this diversity through subreddits, which represent communities that users can join and engage in discussions and share content within specific topics, making Reddit a profuse resource of textual data for developers and researchers looking to collect and analyze. The dataset for this study was collected using Python Reddit API Wrapper (PRAW), a Python package that enables seamless interaction with Reddit data.

Figure 1 illustrates the study's protocol,

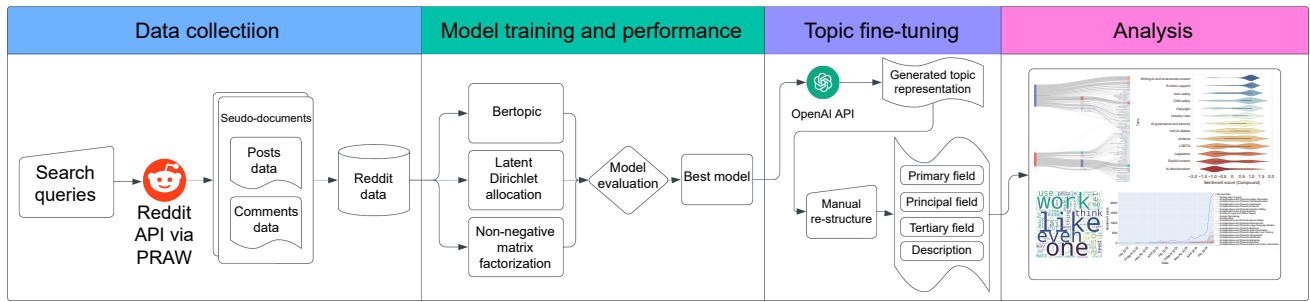


Fig. 1 Protocol of study.

encompassing four phases, which are data collection, model training and performance, topic fine-tuning, and analysis. To begin, a manual search on Reddit was conducted using the keyword “AI” to gather relevant subreddit groups. During this step, irrelevant subreddits were excluded, such as those containing “AI” in unrelated contexts (e.g., “Airbnb”), as well as subreddits with sensitive content, such as those related to explicit material. Once valid subreddits were identified, PRAW was utilized to scrape post data, including post IDs, creation dates, and post texts. The collected post data were then used to retrieve the associated comments, including their text and creation date. To protect users’ privacy, identification information such as usernames is not collected. To process the data, each post and its corresponding comments were combined into a single pseudo-document, which was achieved by using the Python package Pandas.

In our study, over 29 879 posts and nts were collected from 41 distinct subreddits. A sample of post data could be found in Table 2. Attributes of posts include post ID, title, self-text, and created time. Additionally, the posts’ text was aggregated by title and self-text, forming a complete representation of the post. A sample of comments data could be found in

Table 3, where related post ID, body text, and created time were collected. In addition, only root comments, which were directly attached to the posts, were collected during this process. Although a fixed number of 1000 was used during the post collection, PRAW did not always respond with the same number of posts. As a result, the distribution of post numbers across subreddit groups was not even. The detailed number could be found in Table 4. Moreover, although the earliest post obtained was created on 2012-11-09, the majority of the data was from 2023 to 2024, respectively 10 percent and 87 percent of data.

After posts and comments were aggregated into pseudo-documents, a word cloud was generated for preliminary analysis. As shown in Fig. 2, the word cloud highlighted the most frequent terms in the pseudo-documents, such as “work”, “like”, “ai”, “think”, “even”, and “one”, which appeared to align with the recent proliferation of bountiful advanced AI tools. There were a large number of mentions of action, such as “use”, “get”, “make”, and “know”, which indicated the increasing discussion around the pragmatic usage of AI. Also, the presence of words like “issue” and “problem” suggested the challenges and obstacles that users encounter, whether from an active usage standpoint, a perspective of acceptance, or even a passive experience.

Table 2 Sample of the post dataset.

ID	Subreddit	Created	Post text
lf24fyv	ClaudeAI	2024-08-27 00:58:41	Can Claude transcribe a lecture and make it in...
lf8wpzt	StableDiffusion	2024-09-04 15:49:15	RAYGUN, BUT ACTUALLY GOOD[https://youtu.be/Skj...
lf69atw	ChatGPT	2024-09-01 08:02:13	When asked “Name a card, any card”, Claude fav...

Table 3 Sample of the comments dataset.

Post ID	Comment body	Created
l5bol57	Thank the lord. AI hasn’t done drugs.	2023-07-28 17:18:45
l1e085k	I gave the character two wives. My character a...	2023-02-28 18:43:21
le0wkjh	Won’t even need that many bots. The future of...	2024-07-11 20:22:53

3.3.1 Topic diversity

Topic diversity measured how unique each topic is based on top N words.

$$\text{Topic Diversity} = \frac{|\bigcup_{k=1}^K W_k|}{K \times N} \quad (3)$$

where K represents the number of topics; W_k represents the set of the top N words in topic k ; and $|\bigcup_{k=1}^K W_k|$ represents the number of unique words across all topics.

3.3.2 Coherence

The C_v coherence measure was defined as

$$C_v = \frac{1}{M} \sum_{m=1}^M \text{cosine}(v(w_m), v(W_m)) \quad (4)$$

where M represents the number of top words considered for each topic; w_m represents a word in the top M words; W_m represents the set of context words for w_m ; $v(w_m)$ and $v(W_m)$ represent the word vectors of w_m and the context words, respectively; and $\text{cosine}(v(w_m), v(W_m))$ represents the cosine similarity between the word vector $v(w_m)$ and the average word vector of its context words $v(W_m)$.

3.3.3 KL uniform

To access the information gained in resulting topics, Kullback-Leibler (KL) divergence was chosen, which is defined as

$$D_{KL}(P \parallel U) = \sum_{i=1}^n P(i) \log \left(\frac{P(i)}{U(i)} \right) \quad (5)$$

where $P(i)$ represents the probability of the i -th element under the given distribution P ; $U(i) = \frac{1}{n}$ represents the uniform probability for the i -th element, assuming a uniform distribution over n elements; and n represents the number of elements in the distribution.

3.4 Topic fine-tuning and further analysis

After the most suitable model was confirmed, topic fine-tuning could be applied through a semi-AI approach. While topic tuning did not impact the model's performance metrics, it could significantly enhance the interpretability and readability of the output. In other words, transforming a list of keywords into a brief, structured, and concrete sentence provided a clear-cut sense of topic representation. Additionally, this approach reduced the subjective-driven bias that might arise from a straightforward manual label. As the author of BERTopic pointed out, various methods can be employed, including representation-model

based, LLM-based, or a combination of both. In this study, GPT-4o from OpenAI was selected to perform this task via OpenAI's API. Once the new topics were generated, they were further refined through manual labeling, as shown in Fig. 1.

For example, a topic with keywords "comma, car, openpilot, lane, 3x, driving, sunnypilot, cruise, steering, pilot" became "openpilot: open-source autonomous driving and lane assistance system". After additional manual adjustments, it was labeled as "AI Applications and Products, Automation, Autonomous Driving, openpilot, and Lane Assistance System". As shown, topic representation now consisted of four components: primary fields, principal fields, tertiary fields, and a brief description. Although manual labeling was applied, personal bias was minimized, as GPT-generated topics were only corrected if they did not align with their associated documents. This situation was particularly common in AI art topics. For instance, GPT-4o initially generated "Names list: Possibly characters or popular baby names" for a topic, which could be misclassified as cultural. However, the topic indeed related to AI-generated baby pictures, where users provided short descriptions with fictional names.

4 Result

In this section, two topics are discussed, model comparison and analysis of the best model. For model selection, optimized LDA, optimized NMF and Bertopic are compared based on their score across topic diversity, coherence, and KL uniform. Additionally, the result of the best model is chosen to compare the results in other studies within the same benchmark. Then an analysis of the post-topic-tuning model is performed via multiple methods, including a topic representation table, word cloud, Sankey diagram dynamic topic modeling, and sentiment analysis. By applying these techniques, the data are analyzed from aspects of composition, time, and emotion. Furthermore, user behavior and broader implications are discussed.

4.1 Model comparison

Figure 3 illustrates the performance of selected models in each metric. As the plot shows, Bertopic eclipses LDA and NMF in both topic diversity and coherence. However, NMF tied with Bertopic with a difference of 0.01 whilst LDA achieved a low score of

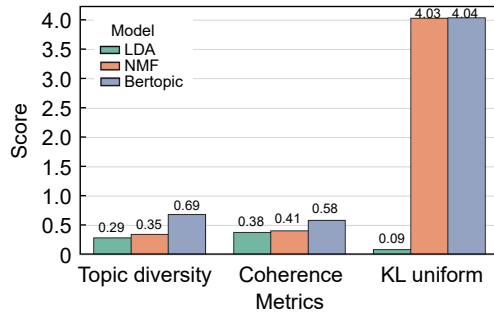


Fig. 3 Comprehensive comparison of the selected models in terms of topic diversity, coherence, and KL uniform.

0.09. In other words, the topics generated by NMF are as informative as the result of Bertopic, and LDA produced a relatively meaningless result. Taken together, the topics yielded by Bertopic are the most unique, consistent, and illuminating. Besides the metrics, it is noteworthy as NMF achieves the same KL uniform score as Bertopic, its file size is only one-third of Bertopic. Therefore, while Bertopic reaches a higher score, it is also more expensive. To further assess the robustness of Bertopic model, other studies' results are compared. In order to compare the coherence score with the same measurement, studies that also used coherence C_v are selected, whereas those used NPMI or Umass are not. As Table 5 suggests, our study has higher coherence than others. While it does not mean other studies are inferior, it shows that our result is acceptable.

4.2 Topic fine tuning and analysis

As the previous section indicated, Bertopic is selected for topic-tuning. Table 6 explicates and reveals the

Table 5 Result comparison with other studies.

Study	Model	Coherence
Our study	BERTopic	0.58
[34]	LDA	0.56
[35]	LDA	0.52
[36]	LDA	0.426
[37]	LDA	0.34

effectiveness of topic-tuning, where “Name” represents the original topic name, and “CustomName” represents the tuned topic name via the semi-AI approach. For topic –1, the name changed from “–1 AI like just people” to “Other, Outlier, Outlier” because topic –1 means outliers in a Bertopic model, and this topic will be also excluded in the later analysis. While the representation, i.e., keywords of topics are limited in Table 6, a word cloud can further assist in the understanding of topic-tuning. Figure 4 displays the word cloud of top 4 most frequent topics in the Bertopic model, and excludes the outlier topic. For the topic (a), the tuned topic presented as “Society, User Support, Technical-related, ChatGPT Tech Support”, which aligns with the word cloud, where the words such as “chatgpt”, “prompt”, “reply”, “question”, and “contact” are frequently used.

The topic (c) resembles with the former, but the shifted the focus from Chatgpt to Google, represented as “Society, User Support, Technical-related, Issues Regard Google Assistant and Voice Interface Technology”. Although its word cloud shows that although they share the same words, such as “chatgpt”, “ai” and “prompt”, the topic (c) also includes words like “gemini” and “google”. Although the topic (b) is

Table 6 Topic-document table.

Topic	Name	Custom name	Representation	Representative document
–1	–1_ai_like_just_people	Other, Outlier, Outlier	AI, like, just, people, im, use, dont, think,...	Companies are going to ai themselves out of...
0	0_flux_lora_loras_forge	Society, User Support, Model-related, ComfyUI Workflow, and Model Training	Flux, lora, loras, forge, images, training,...	I will train a flux lora for you, for free j3...
1	1_reply_conversation_image_concerns	Society, User Support, Technical-related, ChatGPT Tech Support	Reply, conversation, image, concerns, post,...	What? hey /u/ok_leopard_3178! \n\n if your pos...
2	2_gemini_cauchy_google_assistant	Society, User Support, Technical-related, Issues Regarding Google Assistant and Voice Interface Technology	Gemini, cauchy, google, assistant, bor, I5, a...	My gemini ai got this wrong, can someone exp...
...	...	...	...	...



Fig. 4 Top 4 most frequent topic word cloud.

also classified as User Support, it is more Model-related than Technical-related since its wordcloud contains word like “lora”, “training”, “model”, and “image”, which are highly correlated to model training in AI image generation model. Therefore, its description is “ComfyUI Workflow and Model Training”. Compared to (a), (b), and (c), topic (d) is more general because its keywords contain “question”, “post”, “ai”, and “aiart”, which seems to be more platform-related.

For each topic, primary field, principal field, tertiary field, and a brief description were manually assigned. Figure 5 illustrates the classification of topics based on their primary field, principal field, and tertiary field. Two primary fields are identified, which are AI Applications and Products, and Society, whereas the former mainly focus on how AI can be applied to different areas, and the latter not only includes public discussions around AI but also other non-AI topics. There are 29 unique principal fields, where a part of

them can be further broken down into tertiary fields, totaling 65 in all.

AI Applications and Products encompass the diverse use of AI across various fields, with Generative AI (GenAI) being particularly prominent. Although not a single principal field, multiple areas within GenAI, such as development, game design, conversational AI, image generation, large language models, video generation, and writing tools, have gained significant attention. Among these, image generation stands out, with Reddit users primarily employing it for art creation, though some also use it for social media decoration. Another popular area is Conversational AI, which has another name, AI chatbot. A significant number of associate documents within Conversational AI are from platforms like Character AI, which allows users to interact with pre-trained AI chatbots that assimilate different identities, such as real-world celebrities or fictional characters. Additionally, a smaller group of users utilizes conversational AI for emotional support.

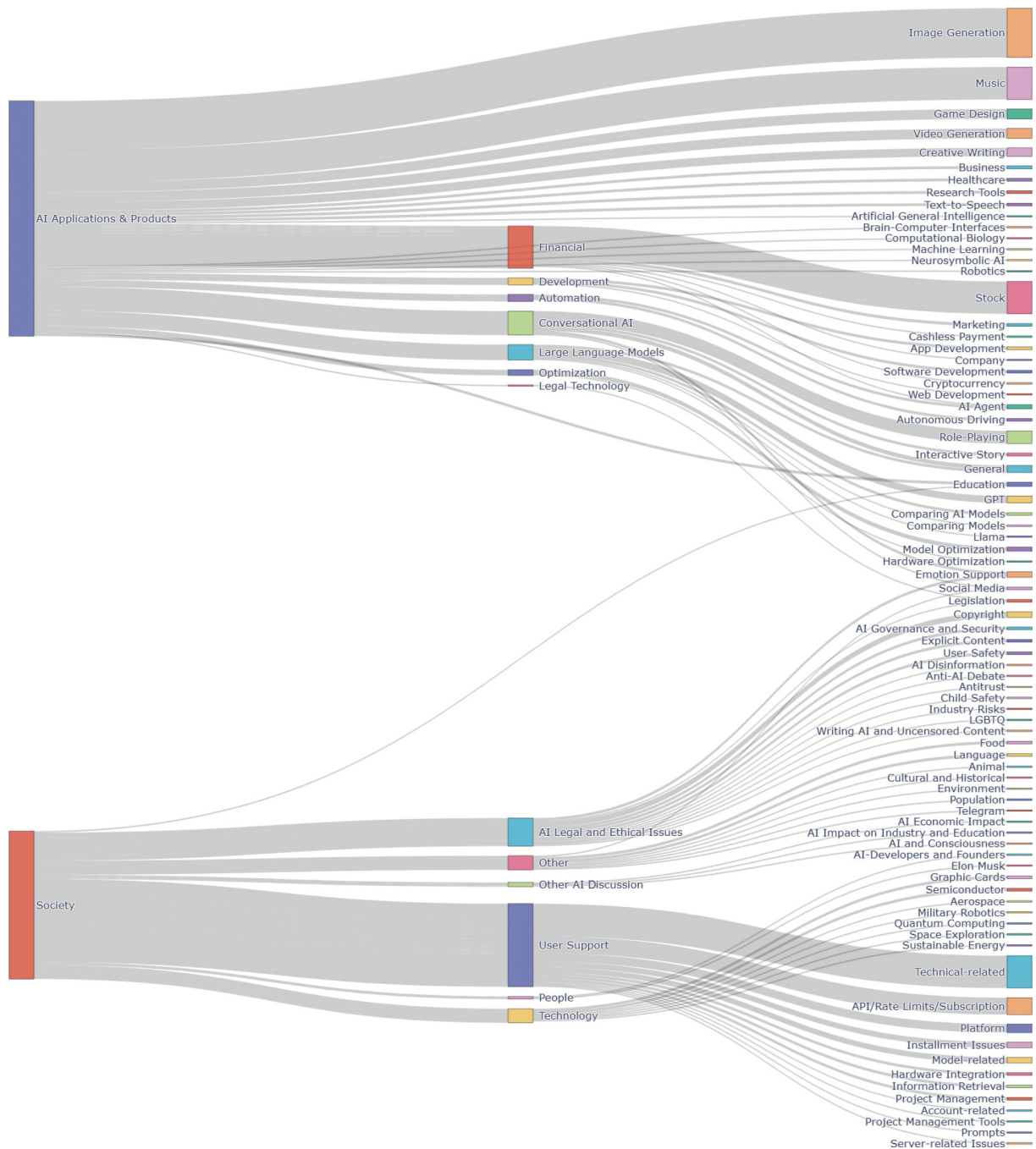


Fig. 5 Classification of Bertopic result.

Outside of GenAI, financial applications also gain interest, particularly in stock prediction, company earnings analysis, and data-driven marketing strategies. Moreover, AI can be applied in Automation, such as Autonomous Driving and AI Agents, offering a more efficient and scalable for industries.

Society can be categorized into six main areas: AI Legal and Ethical Issues, User Support, Other AI Discussions, People, Technology, and Other. User

Support is particularly significant, covering content such as issues encountered by users and advice shared within the community on available tools or platforms. A tectonic portion of this support is technical-related, which involves addressing problems such as recovering chatbot histories, resolving bugs, and handling account-related issues. Subscription of AI Applications and Products, rate limits, and Application Programming Interface (API) issues are also widely discussed by

users who seek to enhance their experiences through paid services. AI Legal and Ethical Issues are another crucial area within Society, where users voiced concerns such as copyright risks, security issues, explicit content generation, and misinformation. In addition, while not necessarily about the drawbacks of AI, AI impacts on the economy, industry, and education are discussed. Moreover, future technologies such as quantum computing, semiconductors, and space exploration, also capture users' attention. Lastly, people like Elon Musk, are mentioned.

GenAI's robustness in generating manuscripts, images, videos, and code has unleashed productivity to the next level, enabling people to create without necessarily learning the corresponding knowledge. While AI applications and Products also contribute to fields like healthcare and finance, where AI serves as a predictor rather than a generator, the use of GenAI appears dominant on Reddit. GenAI shifts the focus from knowledge itself—such as learning how to paint, create melodies, or write a poem—to simply understanding how to use its user interface. This phenomenon explains why User Support is the most-discussed topic within society, and consequently, technical issues are widely discussed. Users encounter diverse degrees of difficulty across different products. For instance, they might struggle with installing models in their local environment or encounter bugs when using the product. Nevertheless, technical knowledge is still required when users want to integrate AI into their products or when the demand for customization exceeds what the original product can offer. The rise of AI-related legal and ethical issues highlights the necessity of making AI products safer and more reliable.

Without proper restrictions, AI could potentially aid malicious activities, making them harder to bulwark.

To access the result with respect to time, Dynamic Topic Modeling (DTM) is employed. Figure 6 visualizes the frequency of topics in a timeline. A description of each topic can be found on the right side of Fig. 6, ordered by the topic frequency. Figure 6 reveals an intriguing pattern of the increased number of AI-related discussions. From July 2022 to July 2024, a diminutive increment can be observed and it gradually decreased with a similar rate from July 2023 to late November 2023. From that point onward, the post frequency started to increase significantly, especially when the time approached March 2024, an explosion of post numbers occurred. Generally, while not all the data points strictly adhered to the overarching trend, the majority of the changing topics were congruent with the trend. When the overall trend was increasing, the changing data points also exhibited an upward trajectory, and when the broader trend was decreasing, the changing topics correspondingly declined. Among all the topics, seven of them exhibit a pronounced accretion, where they are encompassed with the User Support topics and a variety of applications of AI.

Rather than focusing on specific topics, Fig. 7 grouped topics based on their assigned primary field and principal field. User Support appeared in a dominant position with over 2000 documents, meanwhile, other groups have approximately 500 documents or less. Following User Support, a series of subfields from AI Applications and Products emerge, such as image generation and Music. After that, the remaining branches within Society follow, including areas such as Technology and Legal and Ethical Issues. Subsequently, the remaining subfields within AI

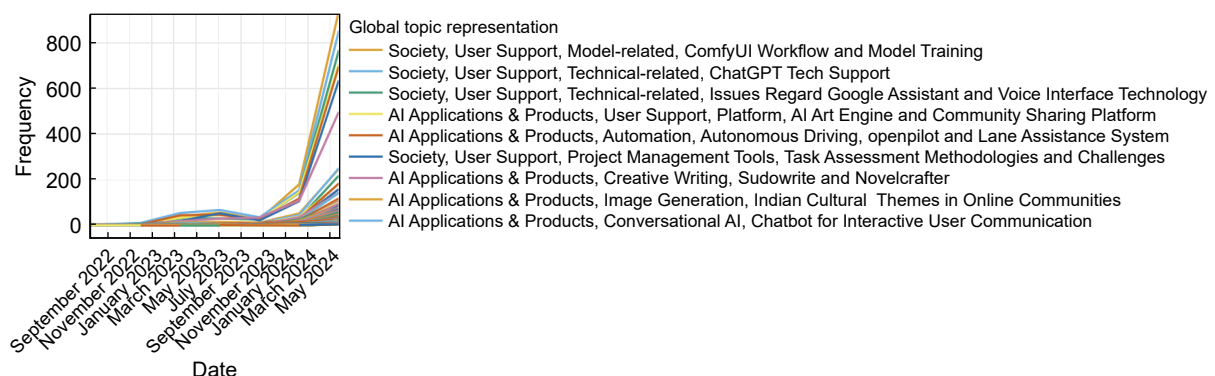


Fig. 6 Dynamic topic modeling of Bertopic results.

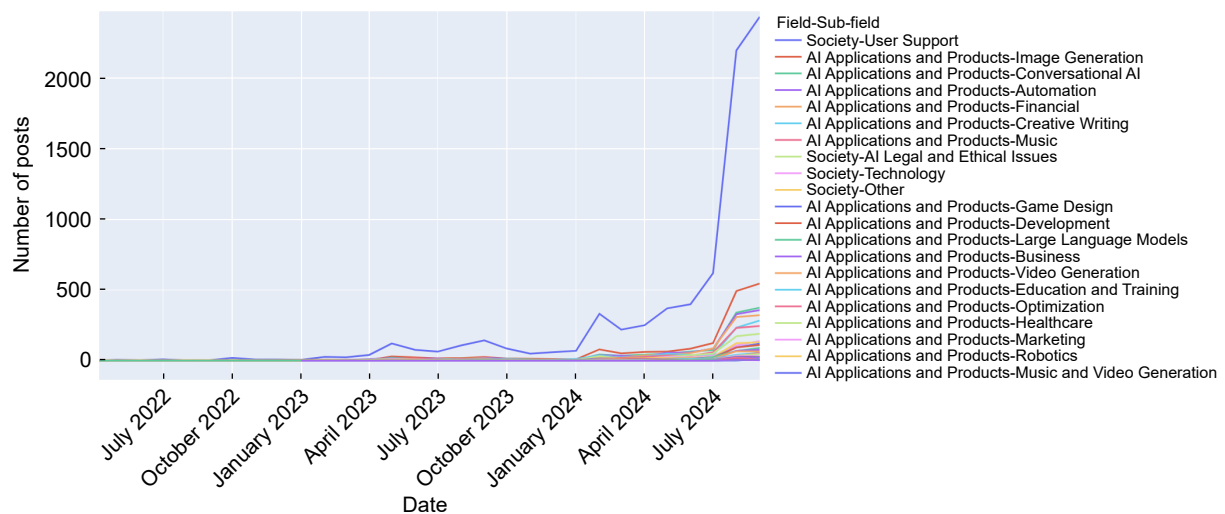


Fig. 7 Primary field-principal field wise dynamic topic modeling plot.

Applications and Products follow, such as Game Design and Robotics. Within the AI Applications and Products, GenAI was found more appealing. For instance, GenAI techniques such as Image Generation, Conversational AI, Creative Writing, and Music have more documents than non-GenAI techniques such as AI Healthcare, Business, and Marketing. Moreover, the substantial lead of User Support infers that using AI requires a certain amount of threshold that many users encounter different issues while using it. Last but not least, the number of AI Legal and Ethical Issues is similar to other AI techniques, implying that public concern against AI also increases as multifarious AI techniques emerge.

Examining the frequency of topics over time is valuable, yet investigating the emotions they convey provides a more multidimensional and insightful perspective. Table 7 presents the sentiment statistics of topics, grouped by primary field and principal field. Across these groups, most mean sentiment values range from 0.2 to 0.5. However, the average sentiment in the “People” field within Society is notably low, at -0.216 , with a high standard deviation suggesting that a specific third field may be contributing to this negativity. While the tertiary fields are not shown in Table 7, further investigation reveals that documents related to Elon Musk received a large amount of negative sentiment, particularly regarding controversies surrounding Starlink.

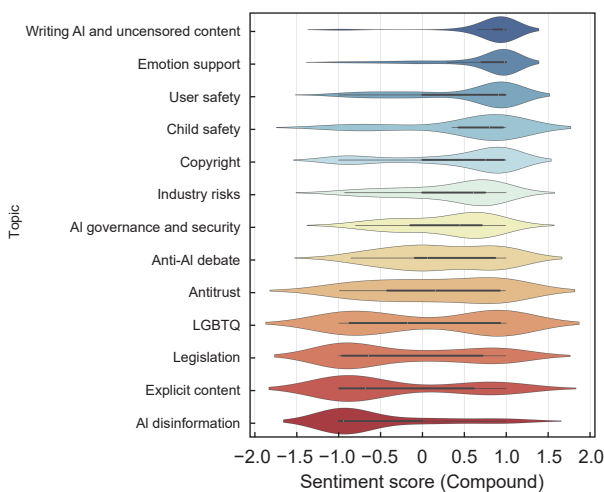
On the other hand, AI algorithms combined with Brain-Computer Interfaces garnered high acclaim,

achieving the highest mean and second-highest median, distinguishing it from other future technology fields such as Computational Biology, Neurosymbolic AI, and Technology, which tended to receive more negative reviews. GenAI-related fields, such as video generation, image generation, creative writing, and text-to-speech, generally received positive sentiment, with the exception of music generation, which had a lower mean sentiment of only 0.272. For non-GenAI technologies, a contrasting trend was observed, that principal fields like Robotics and Artificial General Intelligence received higher average sentiment, whereas Business and Machine Learning had lower scores, suggesting that users may be more enthusiastic about AI in emerging fields rather than traditional ones. As two of the most frequent principal fields within Society, both Technology and Education have positive mean sentiments but also high standard deviations around 0.7, hinting diverse opinions are possibly due to varying experiences within third fields related to these principal fields. Overall, AI Applications and Products reveal specialized fields with distinct sentiment patterns, while Society encompasses more polarized discussions, underscoring the complex and varied emotional responses that AI elicits across different contexts.

To further examine the impact AI brings to society, the tertiary fields within AI Legal and Ethical Issues and User Support are selected for sentiment analysis. Figure 8 shows the sentiment distribution of topics within AI Legal and Ethical Issues, ordered by mean value.

Table 7 Summary statistics of sentiments by primary field and principal field.

Primary field	Principal field	Mean	Median	Standard deviation	Min	Max	Count
AI applications and products	Brain-computer interfaces	0.789	0.988	0.464	−0.894	0.999	37
	Business	0.214	0.296	0.195	−0.612	0.945	249
	Legal technology	0.272	0.128	0.306	−0.459	0.998	44
	Music	0.283	0.077	0.489	−0.975	1.000	803
	Development	0.305	0.500	0.640	−0.999	1.000	337
	Computational biology	0.338	0.361	0.454	−0.494	0.999	29
	Financial	0.347	0.202	0.454	−0.983	0.999	1163
	Healthcare	0.367	0.863	0.784	−1.000	1.000	121
	Video generation	0.441	0.494	0.496	−0.946	1.000	336
	Image generation	0.500	0.700	0.527	−0.999	1.000	1793
	Machine learning	0.138	0.000	0.351	−0.669	0.996	86
	Neurosymbolic AI	0.519	0.562	0.408	−0.176	0.988	24
	Education	0.506	0.664	0.440	−0.868	0.999	168
	Large language models	0.539	0.859	0.551	−0.992	1.000	268
	Research tools	0.758	0.976	0.494	−0.993	0.999	29
	Text-to-speech	0.680	0.922	0.463	−0.993	0.999	107
	Creative writing	0.676	0.925	0.475	−0.997	1.000	876
	Robotics	0.632	0.962	0.631	−1.000	1.000	90
	Conversational AI	0.520	0.857	0.622	−1.000	1.000	1184
	Artificial general intelligence	0.617	0.962	0.547	−0.779	1.000	23
	Automation	0.550	0.856	0.581	−0.998	1.000	1186
	Game design	0.542	0.909	0.628	−1.000	1.000	365
	Optimization	0.568	0.790	0.465	−0.718	1.000	148
Society	Technology	0.561	0.954	0.642	−1.000	1.000	447
	AI legal and ethical issues	0.325	0.720	0.755	−1.000	1.000	638
	Other AI discussion	0.503	0.928	0.679	−0.998	1.000	75
	User support	0.503	0.816	0.604	−1.000	1.000	7930
	Other	0.214	0.000	0.629	−1.000	1.000	465
	People	−0.216	−0.457	0.808	−1.000	0.999	46

**Fig. 8** Sentiment distribution of topics within AI legal and ethical issues.

A polarized pattern can be observed for Explicit-Content-related fields, where some users express a positive attitude toward using AI to generate explicit content, such as uncensored manuscripts, while others react negatively. For safety-related tertiary fields like User Safety and Child Safety, which are closely associated with explicit content but not limited to it (e.g., child pornography and privacy leaks), the left-skewed shape indicates that users tend to neglect these issues when using AI. In contrast, a negative trend is noticed toward AI-generated fraudulent information, where topics like Antitrust and AI Disinformation show lower sentiment and the latter has the lowest median sentiment, close to -1 . Furthermore, discussions around AI Legislation have catalyzed a wave of hostility, as the advancement of AI

legislation is likely to levy more restrictions than currently exist, potentially affecting the use of GenAI, which many users favor. Regarding GenAI, the creation of art has experienced altercation. The sentiment distribution of field Copyright shows a slightly positive trend with a mean of 0.38, which could possibly be due to a larger number of non-artists on Reddit using GenAI to create art, who may feel less concerned about copyright infringement. Although AI poses potential risks to society in multiple ways, the positively skewed distribution of Emotion Support suggests that AI applications, such as Conversational AI, can assist in supporting mental health recovery.

User experience with AI Applications and Products is a crucial benchmark of user satisfaction with AI. Figure 9 shows the sentiment distribution of topics within User Support, ordered by mean value. Compared to AI Legal and Ethical Issues, the tertiary fields within User Support generally receive more positive views from Reddit users, with mean sentiment scores ranging from 0.1 to 0.72. It appears that the difficulty level of tasks correlates with the sentiment they receive. For instance, Prompts and Platform show predominantly positive sentiment, as these discussions often involve sharing prompts that enable GenAI to produce more customized results or recommending useful platforms like Google Colab. Conversely, sentiment is slightly lower for fields such as Technical-related discussions and Hardware Integration, which require more specialized skills and AI knowledge.

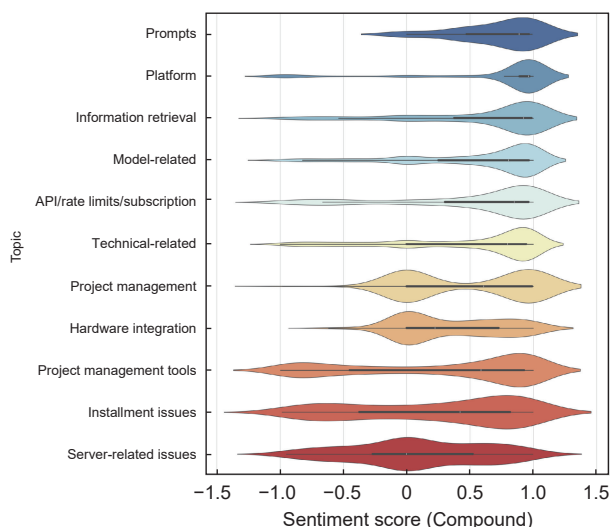


Fig. 9 Sentiment distribution of topics within User Support.

However, the difficulty being discussed here is not merely the technical complexity of a task, but rather a combination of complexity and the frustration it begets.

For instance, Installment Issues might not require high technical skills but can still cause a lot of frustration, as they prevent users from getting started up and engaging with the AI tools. This frustration is often due to external limitations, such as software requirements or dependency conflicts. For example, in one topic that is about “PyTorch Installation Issues with GPU Acceleration and Dependencies”, users express their frustration with Pytorch when installing Stable Diffusion to their local work device. Similarly, server-related issues also evoke low sentiment because problems like product downtime or maintenance are outside of users’ control, thus causing frustration. Therefore, while more complex tasks generally imply lower sentiment, factors such as accessibility and usability barriers also play a critical role in shaping users’ sentiment around AI Applications and Products. On the other hand, factors like the customization of the AI usage experience—as shown by the higher sentiment in topics such as Prompts, Platform, and API/Rate Limits/Subscription—are more likely to increase user satisfaction.

In summary, since Bertopic had the best performance, each of its topic was fine-tuned into four fields. Then, our study applied various techniques to analyze the fine-tuned topics from aspects of composition, time, and sentiment. The results first showed that AI empowered people to produce professional results without traditional training, enabling GenAI a favored choice for Reddit users. Secondly, the discussions around GenAI have increased more than AI in traditional areas such as Business and Healthcare over time, showing a focus on GenAI for AI online. Thirdly, despite the increasing development of AI, the need of restrictions on AI also rises, due to numerous AI legal and ethical issues, where both composition and dynamic topic modeling analysis indicate this urgent concern. Last but not least, task complexity and barriers to usage of AI are two factors that negatively affect user sentiment towards AI while customization of AI has a positive effect.

5 Conclusion

Reddit provides valuable data for topic modeling and

sentiment analysis. This study collected posts and comments data from 41 AI-related subreddits and trained LDA, NMF, and Bertopic with this data. Bertopic stands out due to its higher performance on topic diversity, coherence, and KL uniform. After fine-tuning the topic representation of Bertopic through a semi-AI approach, a high-dimensional analysis was conducted via methods such as Sankey diagram, dynamic topic modeling, and sentiment analysis. Although AI facilitates both creation in traditional and emerging areas, the analysis showed that more actions are needed to protect society from legal and ethical issues caused by AI. This protection should be established not only from the aspect of legislation but also pay more emphasis on education to popularize AI safety, as online users tend to ignore these threats. Moreover, combined with the fact that customization is preferred while complexity is less favored, it is foreseeable that a safe, customized, decentralized, and user-friendly AI will become more prevalent in the future. In addition, although AI in traditional area receive less attention than in emerging areas, it is likely that traditional area attracts more focus as more advanced technology is being implemented in various aspects of society. There are several directions for future research to enhance this study. While this study provides reasonable conjectures about topic trends over time and sentiment, empirical approaches, such as statistical testing, could produce more reliable insights. Furthermore, in addition to collecting more data from Reddit, a cross-platform study incorporating other social platforms such as Twitter, could provide a broader perspective.

Conflict of Interest

The authors declare no conflict of interest.

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